Rational and irrational numbers

MOPSS

Mathematics Olympiad

Problem Solving Sessions

Department of Mathematics
IISER Bhopal

https://jpsaha.github.io/MOTP/MOPSS/

Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

§1 Rational and irrational numbers

Exercise 1.1 (British Mathematical Olympiad Round 2 2025 P1)**.** Prove that if n is a positive integer, then $\frac{1}{n}$ can be expressed as a finite sum of reciprocals of different triangular numbers. (A **triangular number** is an integer which is equal to $\frac{k(k+1)}{2}$ for some positive integer k.)

Walkthrough —

(a) Note that it suffices to express $\frac{1}{2n}$ as a finite sum of reciprocal of consecutive integers.

(b) Observe that
$$\frac{1}{2n} = \frac{1}{n} - \frac{1}{2n}$$
holds.

Solution 1. Note that
$$\frac{1}{2n} = \left(\frac{1}{n} - \frac{1}{n+1}\right) + \left(\frac{1}{n+1} - \frac{1}{n+2}\right) + \cdots + \left(\frac{1}{2n+1} - \frac{1}{2n}\right)$$
holds. This yields that $\frac{1}{n}$ is the sum of the reciprocals of
$$\frac{n(n+1)}{2}, \frac{(n+1)(n+2)}{2}, \ldots, \frac{(2n+1)2n}{2},$$
which are distinct triangular numbers since the map $x \mapsto \frac{x(x+1)}{2}$ from $\mathbb{R} \to \mathbb{R}$ is injective. ■

Exercise 1.2 (Brazil National Olympiad 2020 Level 3 P1, AoPS)**.** Prove that there are positive integers $a_1, a_2, \ldots, a_{2020}$ such that
$$\frac{1}{a_1} + \frac{1}{2a_2} + \frac{1}{3a_3} + \cdots + \frac{1}{2020a_{2020}} = 1.$$

Walkthrough —

(a) Show that for any positive integer n, there exist positive integers $a_1, a_2, \ldots, a_n$ such that
$$\frac{1}{a_1} + \frac{1}{2a_2} + \frac{1}{3a_3} + \cdots + \frac{1}{na_n} = 1.$$

(b) Observe that $1 = \frac{1}{2} + \frac{1}{2}$, $1 = \frac{1}{2} + \frac{1}{3} + \frac{1}{6}$, $1 = \frac{1}{2} + \frac{1}{6} + \frac{1}{4} + \frac{1}{12}$ etc. hold.

(c) Also note that
$$\frac{1}{n} = \frac{1}{n+1} + \frac{1}{n(n+1)}.$$

Solution 2. We claim that for any positive integer n, there exist positive integers $a_1, a_2, \ldots, a_n$ such that
$$\frac{1}{a_1} + \frac{1}{2a_2} + \frac{1}{3a_3} + \cdots + \frac{1}{na_n} = 1.$$
We will prove this claim by induction on n. The base case $n = 1$ is trivial, since we can take $a_1 = 1$. Also note that the case $n = 2$ holds, since we can take $a_1 = 2$ and $a_2 = 1$. Assume now that the claim holds for some $n \geq 2$. Hence, there exist positive integers $a_1, a_2, \ldots, a_n$ such that
$$\frac{1}{a_1} + \frac{1}{2a_2} + \frac{1}{3a_3} + \cdots + \frac{1}{na_n} = 1.$$
Using
$$\frac{1}{n} = \frac{1}{n+1} + \frac{1}{n(n+1)},$$
we obtain
$$\begin{aligned} 1 &= \frac{1}{a_1} + \frac{1}{2a_2} + \frac{1}{3a_3} + \cdots + \frac{1}{na_n} \\ &= \frac{1}{a_1} + \frac{1}{2a_2} + \frac{1}{3a_3} + \cdots + \frac{1}{n(n+1)a_n} + \frac{1}{(n+1)a_n}. \end{aligned}$$
This shows that the claim also holds for $n+1$. By the principle of mathematical induction, the claim holds for all positive integers n. In particular, the claim holds for $n = 2020$, which completes the proof. ■

Solution 3. Note that
$$\frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{n(n+1)} = 1 - \frac{1}{n+1}$$
holds for all positive integers n. Taking
$$a_1 = n+1, a_2 = 1, a_3 = 2, \ldots, a_{n+1} = n$$

yields

$$\frac{1}{a_1} + \frac{1}{2a_2} + \frac{1}{3a_3} + \cdots + \frac{1}{(n+1)a_{n+1}} = 1.$$

■

Example 1.3 (Putnam 2009 B1)**.** Show that every positive rational number can be written as a quotient of products of factorials of (not necessarily distinct) primes. For example,

$$\frac{10}{9} = \frac{2! \cdot 5!}{3! \cdot 3! \cdot 3!}.$$

Example 1.4. Let $n \geq 2$ be an integer. Let m be the largest integer which is less than or equal to n, and which is a power of 2. Put $\ell_n =$ the least common multiple of $1, 2, \ldots, n$. Show that l_n/m is odd, and that for every integer $k \leq n, k \neq m, l_n/k$ is even. Hence prove that

$$1 + \frac{1}{2} + \cdots + \frac{1}{n}$$

is not an integer.

Example 1.5 (Moscow Math Circles)**.** Does there exist irrational numbers x, y with $x > 0$ such that x^y is rational?

Summary — Consider $\sqrt{2}^{\sqrt{2}}$.

Walkthrough —

(a) Consider $\sqrt{2}^{\sqrt{2}}$.

(b) If $\sqrt{2}^{\sqrt{2}}$ is rational, then we are done by taking $x = y = \sqrt{2}$.

(c) If $\sqrt{2}^{\sqrt{2}}$ is irrational, then can you find out suitable x, y?

Solution 4. Consider $\sqrt{2}^{\sqrt{2}}$. If $\sqrt{2}^{\sqrt{2}}$ is rational, then we may take $x = y = \sqrt{2}$. If $\sqrt{2}^{\sqrt{2}}$ is irrational, then taking $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$, we find that

$$x^y = \left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = \left(\sqrt{2}\right)^2 = 2,$$

which is a rational number. ■

Example 1.6 (India RMO 2013d P4)**.** Let x be a nonzero real number such that $x^4 + \frac{1}{x^4}$ and $x^5 + \frac{1}{x^5}$ are both rational numbers. Prove that $x + \frac{1}{x}$ is a rational number.

Summary — Consider the difference
$$\left(x^5+\frac{1}{x^5}\right)-\left(x^4+\frac{1}{x^4}\right)\left(x+\frac{1}{x}\right).$$

Solution 5. For a positive integer n, put $y_n = x^n + \frac{1}{x^n}$. Note that
$$y_5 = y_1y_4 - y_3 = y_1y_4 - y_1(y_2-1) = y_1(y_4-y_2+1),$$
and $y_4 - y_2 + 1$ is nonzero, otherwise, we would get $y_2^2 - y_2 - 1 = 0$, that is, $y_2 = \frac{1\pm\sqrt{5}}{2}$, which shows that $y_4 = \frac{3\pm\sqrt{5}}{2}$, which contradicts that y_4 is rational. To show that y_1 is rational, it suffices to show that y_2 is rational. Observe that
$$y_{10} = y_2y_8 - y_6 = y_2y_8 - y_2y_4 + y_2 = y_2(y_8 - y_4 + 1),$$
and $y_8 - y_4 + 1 \neq 0$ (otherwise, we would obtain $y_4^2 - y_4 - 1 = 0$, which contradicts the rationality of y_4). Since y_{10}, y_8, y_4 are rational, it follows that y_2 is rational. Using
$$y_5 = y_1(y_4 - y_2 + 1),$$
and that $y_4 - y_2 + 1$ is nonzero, we conclude that y_1 is rational. ■

Example 1.7 (All-Russian MO 2001–2002 Final stage Grade 11 P1)**.** Real numbers x and y are such that $x^p + y^q$ is rational for any different odd primes p, q. Show that x and y are rational.

Solution 6. The given condition implies that for any three distinct primes p, q, r,
$$x^p - x^q, x^q - x^r$$
are rational since
$$x^p - x^q = (x^p + y^r) - (x^q + y^r), x^q - x^r = (x^q + y^p) - (x^r + y^p).$$
It follows that
$$a = x^7 - x^5, b = x^5 - x^3$$
are rational. If $b = 0$, then $x = 0$ or $x = \pm 1$, and hence x is rational. If $b \neq 0$, then note that $x^2 = \frac{a}{b}$, which shows that x^2 is rational. Observing that
$$b = x^2(x^2-1)x,$$
it follows that if $b \neq 0$, then x is rational. The rationality of y follows similarly. ■

Exercise 1.8 (British Mathematical Olympiad Round 1 2004/5 P5)**.** Let S be a set of rational numbers with the following properties:

1. $\frac{1}{2} \in S$,
2. If $x \in S$, then both $\frac{1}{x+1} \in S$ and $\frac{x}{x+1} \in S$.

Prove that S contains all rational numbers in the interval $0 < x < 1$.

Walkthrough —

(a) Since $\frac{1}{2}$ lies in S, by the second condition, it follows that $\frac{2}{3}$ lies in S and so does $\frac{1}{3}$.

(b) Taking $x = \frac{1}{3}$, it follows that
$$\frac{3}{4}, \frac{1}{4}$$
lie in S. **Note that we have showed that S contains all the rationals between 0 and 1 with denominator at most 4.**

(c) Taking $x = \frac{2}{3}$, it follows that
$$\frac{2}{5}, \frac{3}{5}$$
lie in S. We are **not in a position** to conclude that S contains all the rationals between 0 and 1 with denominator at most 5.

(d) Taking $x = \frac{1}{4}$, it follows that
$$\frac{1}{5}, \frac{4}{5}$$
lie in S. It follows that S contains all the rationals between 0 and 1 with denominator at most 5.

(e) Does the above provide any insight to conclude that S contains all the rationals between 0 and 1? For instance, can one expect the following (and then prove, or realize that it is false, or argue along different lines)?

> For a rational number x lying in S, the rationals
> $$\frac{1}{x+1}, \frac{x}{x+1}$$
> have denominators larger[a] than that of x.
>
> [a]Often, while being naive, one takes the liberty to write **larger** to mean **no smaller**, that is, **greater than or equal to**. But this is **NOT allowed** while writing down a solution.

Or, stated in a different way,

> A rational number lying in $(0, 1)$ can be obtained from a rational number lying in $(0, 1)$ with smaller denominator by applying one of the maps
> $$x \mapsto \frac{1}{x+1}, x \mapsto \frac{x}{x+1}.$$

Solution 7. It suffices to establish the following.

Claim — For any integer $k \geq 2$, all the rationals lying in $(0,1)$ with denominators not exceeding k lie in S, that is, we have

$$\left\{\frac{1}{\ell}, \frac{2}{\ell}, \dots, \frac{\ell-2}{\ell}, \frac{\ell-1}{\ell}\right\} \subseteq S \quad \text{for all } 2 \leq \ell \leq k. \tag{1}$$

Proof of the Claim. Eq. (1) holds for $k = 2$ from condition (1). Suppose Eq. (1) holds for $k = n - 1$ for some integer $n \geq 3$. Let m be an integer satisfying $1 \leq m < n$. Using the induction hypothesis, we will show that $\frac{m}{n}$ lies in S. Note that for $0 < x < 1$, the inequalities

$$0 < \frac{x}{x+1} < \frac{1}{2}, \frac{1}{2} < \frac{1}{x+1} < 1$$

hold. Using Condition (1), it follows that $\frac{m}{n}$ lies in S if $\frac{m}{n} = \frac{1}{2}$. If $0 < \frac{m}{n} < \frac{1}{2}$, then

$$\frac{x}{x+1} = \frac{m}{n}$$

holds for $x = \frac{m}{n-m}$, which is a rational number lying in $(0,1)$ with denominator $\leq n-1$, and by induction hypothesis, the set S contains $\frac{m}{n}$. Moreover, if $\frac{1}{2} < \frac{m}{n} < 1$, then

$$\frac{1}{x+1} = \frac{m}{n}$$

holds for $x = \frac{n-m}{m}$, which is a rational number lying in $(0,1)$ with denominator $\leq n-1$, and by induction hypothesis, the set S contains $\frac{m}{n}$. We conclude that for any integer $n \geq 3$, Eq. (1) holds for $k = n$ if it holds for $k = n - 1$. □

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Example 1.9 (Junior Balkan MO TST 1999)**.** Let S be a set of rational numbers with the following properties:

1. $\frac{1}{2} \in S$,
2. If $x \in S$, then both $\frac{x}{2} \in S$ and $\frac{1}{x+1} \in S$.

Prove that S contains all the rational numbers from the interval $(0,1)$.

Walkthrough —

(a) Taking $x = \frac{1}{2}$, it follows that S contains $\frac{2}{3}$, and hence it also contains $\frac{1}{3}$.

(b) Taking $x = \frac{1}{2}$, it follows that S contains $\frac{1}{4}$. Next, taking $x = \frac{1}{3}$, we obtain that S contains $\frac{3}{4}$.

(c) Applying the map $x \mapsto \frac{1}{x+1}$ to $x = \frac{2}{3}, \frac{1}{4}$, it follows that S contains $\frac{3}{5}, \frac{4}{5}$. Since S contains $\frac{4}{5}$, the set S also contains $\frac{2}{5}, \frac{1}{5}$.

(d) Applying $x \mapsto \frac{1}{x+1}$ to $x = \frac{1}{5}$, it follows that S contains $\frac{5}{6}$. Note that S

contains $\frac{4}{6} = \frac{2}{3}, \frac{3}{6} = \frac{1}{2}, \frac{2}{6} = \frac{1}{3}$. It also follows that S contains $\frac{1}{6}$.

(e) Does the above provide any insight into the problem? Can one expect the following?

> The rationals lying in $(\frac{1}{2}, 1)$ can be obtained by applying the map $x \mapsto \frac{1}{x+1}$ to the rationals lying in $(0, 1)$ with small denominators. Moreover, a rational number r lying in $(0, \frac{1}{2})$ can be obtained by applying the map $x \mapsto \frac{x}{2}$ to the rationals lying in $(\frac{1}{2}, 1)$, more specifically, to those rationals with denominators at most the denominator of r.

Solution 8. It suffices to establish the following.

> **Claim** — For any integer $k \geq 2$, all the rationals lying in $(0, 1)$ with denominators not exceeding k lie in S, that is, we have
>
> $$\left\{\frac{1}{\ell}, \frac{2}{\ell}, \dots, \frac{\ell-2}{\ell}, \frac{\ell-1}{\ell}\right\} \subseteq S \quad \text{for all } 2 \leq \ell \leq k. \tag{2}$$

Proof of the Claim. Note that Eq. (2) holds for $k = 2$ by hypothesis. Suppose Eq. (2) holds for $k = n - 1$ where $n \geq 3$ is an integer. Let m be an integer satisfying $1 \leq m < n$. Using the induction hypothesis, we will show that $\frac{m}{n}$ lies in S. Note that for $0 < x < 1$, the inequalities

$$\frac{1}{2} < \frac{x}{x+1} < 1$$

hold. If $\frac{m}{n}$ is equal to $\frac{1}{2}$, then it lies in S by hypothesis. If $\frac{m}{n}$ lies in $(\frac{1}{2}, 1)$, then

$$\frac{1}{x+1} = \frac{m}{n}$$

holds for $x = \frac{n-m}{m}$, which is a rational number lying in $(0, 1)$ with denominator $\leq n - 1$, and by induction hypothesis, the set S contains $\frac{m}{n}$. If $\frac{m}{n}$ lies in $(0, \frac{1}{2})$, then the rational number $\frac{2m}{n}$ lies in $(\frac{1}{2}, 1)$, and if it has denominator $< n$ (when expressed in its least form), then it is an element of S by the induction hypothesis, and if it has denominator equal to n (when expressed in its least form), then by the above argument, it lies in S, and consequently, S contains $\frac{m}{n}$. We conclude that for any integer $n \geq 3$, Eq. (2) holds for $k = n$ if it holds for $k = n - 1$. □

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Example 1.10 (India RMO 2019a P1)**.** Suppose x is a nonzero real number such that both x^5 and $20x + \frac{19}{x}$ are rational numbers. Prove that x is a rational number.

Solution 9. Let us establish the following more general claim.

Claim — Let x be a nonzero real number and r be a rational number such that x^5 and $x + \frac{r}{x}$ are rational numbers. Then x is a rational number.

Proof of the Claim. Note that

$$x^2 + \frac{r^2}{x^2}, x^3 + \frac{r^3}{x^3}$$

are rational. Write $s = x^5$ and note that

$$x^3 + \frac{r^3}{x^3} = \frac{r^3}{s}x^2 + \frac{s}{x^2}.$$

It follows that there are rational numbers a, b such that

$$\begin{pmatrix} 1 & r^2 \\ \frac{r^3}{s} & s \end{pmatrix} \begin{pmatrix} x^2 \\ \frac{1}{x^2} \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

holds. If the determinant of $\begin{pmatrix} 1 & r^2 \\ \frac{r^3}{s} & s \end{pmatrix}$ is nonzero, then it follows that x^2 is rational, and using the rationality of x^5, we obtain that x is rational. If the determinant of $\begin{pmatrix} 1 & r^2 \\ \frac{r^3}{s} & s \end{pmatrix}$ is zero, then we get $s^2 = r^5$, which implies that $x^{10} = r^5$, and hence, x^2 is rational, and consequently, so is x. □

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References

[Che25] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)